

Cosmology

Physics, Mathematics & Research Frontiers

“The universe is under no obligation to make sense to you.”

— Neil deGrasse Tyson

A Comprehensive Technical Review

Covering: General Relativity · Friedmann Equations · Inflation · Dark Matter
Dark Energy · CMB Physics · Large-Scale Structure · Quantum Cosmology

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This document provides a graduate-level treatment of modern cosmology,
including mathematical derivations and current research directions.

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1 Introduction to Cosmology

Cosmology is the scientific study of the origin, evolution, structure, and ultimate fate of the universe as a whole. It sits at the intersection of general relativity, particle physics, thermodynamics, and quantum field theory. The subject was transformed in the 20th century by three foundational observations:

1. **Hubble's Law** (1929): Galaxies recede with velocity proportional to distance.
2. **Cosmic Microwave Background** (Penzias & Wilson, 1965): Relic thermal radiation at $T \approx 2.725$ K.
3. **Big Bang Nucleosynthesis**: Predicted abundances of light elements match observations.

Together these form the pillars of the **Hot Big Bang model**, now refined into the standard Λ CDM (Lambda Cold Dark Matter) cosmological model.

2 Mathematical Foundations: General Relativity

2.1 Spacetime and the Metric

The central object of general relativity is the **metric tensor** $g_{\mu\nu}$, which encodes the geometry of spacetime. The line element (invariant interval) is:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (1)$$

with the Einstein summation convention over repeated indices $\mu, \nu = 0, 1, 2, 3$.

2.2 The Christoffel Symbols and Geodesics

The Levi-Civita connection coefficients (Christoffel symbols) are:

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) \quad (2)$$

Free-falling particles follow **geodesics**, satisfying:

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0 \quad (3)$$

2.3 Curvature Tensors

The Riemann curvature tensor measures the non-commutativity of covariant derivatives:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda \quad (4)$$

Contracting gives the **Ricci tensor** and **Ricci scalar**:

$$R_{\mu\nu} = R^\lambda_{\mu\lambda\nu}, \quad R = g^{\mu\nu} R_{\mu\nu} \quad (5)$$

The **Einstein tensor** is:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \quad (6)$$

which satisfies the contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$.

2.4 Einstein Field Equations

The **Einstein Field Equations (EFE)** relate geometry to matter-energy:

Einstein Field Equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (7)$$

where Λ is the cosmological constant, G is Newton's gravitational constant, and $T_{\mu\nu}$ is the stress-energy tensor.

These are 10 coupled, nonlinear PDEs. The stress-energy tensor for a perfect fluid is:

$$T_{\mu\nu} = \left(\rho + \frac{p}{c^2} \right) u_\mu u_\nu + p g_{\mu\nu} \quad (8)$$

where ρ is energy density, p is pressure, and u^μ is the four-velocity.

3 The Friedmann-Lemaître-Robertson-Walker Universe

3.1 The FLRW Metric

Assuming the **Cosmological Principle** (homogeneity and isotropy on large scales), the most general metric is the **FLRW metric**:

FLRW Line Element

$$ds^2 = -c^2 dt^2 + a(t)^2 \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] \quad (9)$$

where:

- $a(t)$ is the **scale factor**, normalized $a(t_0) = 1$
- $k = +1, 0, -1$ denotes closed, flat, or open spatial curvature
- $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ is the solid angle element

3.2 Friedmann Equations

Substituting the FLRW metric into the EFE yields the **Friedmann equations**:

Friedmann Equations

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3} \quad (10)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3} \quad (11)$$

Here $H \equiv \dot{a}/a$ is the **Hubble parameter**. The current value $H_0 \approx 67\text{--}73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is subject to the *Hubble tension* (see Section 13.1).

3.3 Fluid Equation and Equation of State

Energy conservation $\nabla_\mu T^{\mu\nu} = 0$ gives the **continuity equation**:

$$\dot{\rho} + 3H\left(\rho + \frac{p}{c^2}\right) = 0 \quad (12)$$

With equation of state $p = w\rho c^2$, this integrates to:

$$\rho \propto a^{-3(1+w)} \quad (13)$$

Component	w	$\rho \propto$
Matter (dust)	0	a^{-3}
Radiation	1/3	a^{-4}
Cosmological constant Λ	-1	a^0 (constant)
Curvature	-1/3	a^{-2}

3.4 Density Parameters and Critical Density

The **critical density** is the energy density for a spatially flat universe:

$$\rho_c(t) = \frac{3H(t)^2}{8\pi G} \quad (14)$$

Dimensionless **density parameters** $\Omega_i \equiv \rho_i/\rho_c$ allow Eq. (10) to be written:

$$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1 \quad (15)$$

where $\Omega_k \equiv -kc^2/(aH)^2$. Observations give:

$$\begin{aligned}\Omega_m &\approx 0.315 && \text{(matter: baryonic + dark)} \\ \Omega_\Lambda &\approx 0.685 && \text{(dark energy)} \\ \Omega_k &\approx 0 && \text{(spatially flat)}\end{aligned}$$

4 Cosmic Expansion and Distances

4.1 Redshift

As photons travel through an expanding universe, their wavelength stretches:

$$1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a(t_0)}{a(t_{\text{em}})} = \frac{1}{a(t_{\text{em}})} \quad (16)$$

4.2 Comoving and Proper Distances

The **comoving distance** to redshift z is:

$$\chi(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad (17)$$

where $E(z) \equiv H(z)/H_0 = \sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda}$.

The **luminosity distance** (used in standard candles):

$$d_L = (1+z) \chi(z) \quad (18)$$

The **angular diameter distance** (used in standard rulers):

$$d_A = \frac{\chi(z)}{1+z} \quad (19)$$

5 Cosmic Inflation

5.1 Motivations: Problems of the Standard Big Bang

The Hot Big Bang model faces several fine-tuning issues:

Problems Solved by Inflation

- **Horizon problem:** Causally disconnected regions of the CMB have the same temperature to 10^{-5} .
- **Flatness problem:** $|\Omega - 1|$ was $< 10^{-60}$ at the Planck epoch to give $|\Omega_0 - 1| <$

0.005 today.

- **Monopole problem:** GUT phase transitions produce magnetic monopoles not observed.

5.2 Inflationary Dynamics

Inflation posits a period of accelerated expansion ($\ddot{a} > 0$) driven by a scalar **inflaton field** ϕ with potential $V(\phi)$.

The action is:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\partial\phi)^2 - V(\phi) \right] \quad (20)$$

The equations of motion in a flat FLRW background:

$$H^2 = \frac{1}{3M_{\text{Pl}}^2} \left[\frac{1}{2} \dot{\phi}^2 + V(\phi) \right] \quad (21)$$

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0 \quad (22)$$

5.3 Slow-Roll Approximation

When $\dot{\phi}^2 \ll V(\phi)$ and $|\ddot{\phi}| \ll 3H|\dot{\phi}|$ (**slow-roll**):

$$H^2 \approx \frac{V}{3M_{\text{Pl}}^2}, \quad 3H\dot{\phi} \approx -V' \quad (23)$$

The slow-roll parameters are:

$$\epsilon_V \equiv \frac{M_{\text{Pl}}^2}{2} \left(\frac{V'}{V} \right)^2 \ll 1, \quad \eta_V \equiv M_{\text{Pl}}^2 \frac{V''}{V} \ll 1 \quad (24)$$

Inflation ends when $\epsilon_V \sim 1$.

5.4 Number of e-Folds

The amount of inflation is characterised by **e-folds**:

$$N \equiv \ln \frac{a_{\text{end}}}{a_*} = \int_{t_*}^{t_{\text{end}}} H dt \approx \frac{1}{M_{\text{Pl}}^2} \int_{\phi_{\text{end}}}^{\phi_*} \frac{V}{V'} d\phi \quad (25)$$

Solving the horizon and flatness problems requires $N \gtrsim 60$.

5.5 Primordial Perturbations

Quantum fluctuations of ϕ during inflation seed all structure. The power spectrum of **scalar (curvature) perturbations** is:

$$\mathcal{P}_{\mathcal{R}}(k) = \frac{H^2}{8\pi^2 M_{\text{Pl}}^2 \epsilon_V} \Big|_{k=aH} \quad (26)$$

Parameterised as a power law:

$$\mathcal{P}_{\mathcal{R}}(k) = A_s \left(\frac{k}{k_*} \right)^{n_s-1} \quad (27)$$

where $A_s \approx 2.1 \times 10^{-9}$ and the **spectral index** $n_s \approx 0.965$ (slightly red-tilted).

Tensor perturbations (primordial gravitational waves) have:

$$\mathcal{P}_T(k) = \frac{2H^2}{\pi^2 M_{\text{Pl}}^2} \Big|_{k=aH} \quad (28)$$

The **tensor-to-scalar ratio** $r \equiv \mathcal{P}_T/\mathcal{P}_{\mathcal{R}} = 16\epsilon_V$ is constrained $r < 0.036$ (Planck+BICEP/Keck 2021).

6 Thermal History of the Universe

6.1 Temperature and Entropy

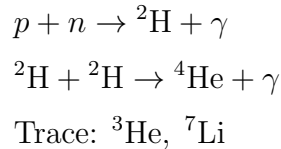
In thermal equilibrium, the photon temperature scales as $T \propto a^{-1}$. The total radiation energy density:

$$\rho_r = \frac{\pi^2}{30} g_*(T) T^4 \quad (29)$$

where $g_*(T)$ counts relativistic degrees of freedom.

6.2 Big Bang Nucleosynthesis (BBN)

At $T \sim 1 \text{ MeV}$ ($t \sim 1 \text{ s}$), weak interactions freeze out, fixing the neutron-to-proton ratio $n/p \approx 1/7$. Between $T \sim 0.1\text{--}0.01 \text{ MeV}$, light nuclei form:



Predicted mass fraction $Y_p \approx 0.247$, consistent with observation.

6.3 Recombination and Photon Decoupling

At $z \approx 1100$ ($T \approx 0.3$ eV), electrons combine with protons:



The ionisation fraction is governed by the **Saha equation**:

$$\frac{x_e^2}{1 - x_e} = \frac{1}{n_b} \left(\frac{m_e T}{2\pi} \right)^{3/2} e^{-B_H/T} \quad (31)$$

where $B_H = 13.6$ eV is the hydrogen binding energy and $x_e = n_e/n_b$.

When $x_e \rightarrow 0$, photons decouple and free-stream — forming the **Cosmic Microwave Background**.

7 The Cosmic Microwave Background

7.1 Temperature Anisotropies

The CMB temperature field is expanded in spherical harmonics:

$$\frac{\Delta T}{T}(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell}^m(\hat{n}) \quad (32)$$

The **angular power spectrum**:

$$C_{\ell} = \frac{1}{2\ell + 1} \sum_m |a_{\ell m}|^2 \quad (33)$$

7.2 Acoustic Oscillations

Before recombination, the photon-baryon fluid undergoes acoustic oscillations. The temperature power spectrum shows characteristic **acoustic peaks** at:

$$\ell_n \approx \frac{n\pi d_A(z_{\text{rec}})}{r_s(z_{\text{rec}})} \quad (34)$$

where r_s is the sound horizon at recombination:

$$r_s = \int_0^{t_{\text{rec}}} \frac{c_s dt}{a(t)}, \quad c_s = \frac{c}{\sqrt{3(1+R)}}, \quad R = \frac{3\rho_b}{4\rho_{\gamma}} \quad (35)$$

7.3 Polarisation

The CMB is partially polarised via Thomson scattering near last scattering. Decomposed into:

- **E-modes:** Curl-free, generated by scalar perturbations
- **B-modes:** Divergence-free, generated by tensor perturbations (primordial GW) or weak lensing

Detecting primordial B-modes would directly probe inflation at energy scale:

$$V^{1/4} \sim 1.06 \times 10^{16} \text{ GeV} \left(\frac{r}{0.01} \right)^{1/4} \quad (36)$$

7.4 Sachs-Wolfe Effect

CMB photons climbing out of potential wells are redshifted. The **Sachs-Wolfe effect** gives:

$$\left. \frac{\Delta T}{T} \right|_{\text{SW}} = \frac{1}{3} \Phi(\mathbf{x}_{\text{dec}}) \quad (37)$$

where Φ is the gravitational potential. The **integrated Sachs-Wolfe (ISW)** effect arises from time-varying potentials along the line of sight:

$$\left. \frac{\Delta T}{T} \right|_{\text{ISW}} = -2 \int_0^{\chi^*} \dot{\Phi} d\chi \quad (38)$$

8 Dark Matter

8.1 Evidence

Evidence for dark matter (DM) operates on multiple scales:

Observational Evidence for Dark Matter

- **Galaxy rotation curves:** Flat $v(r)$ at large r implies $M(< r) \propto r$, inconsistent with visible matter.
- **Gravitational lensing:** Bullet Cluster shows mass offset from baryons.
- **CMB:** Acoustic peak heights and positions constrain $\Omega_b h^2$ and $\Omega_c h^2$ independently.
- **Structure formation:** DM halos provide gravitational seeds for galaxy formation.

8.2 DM Density Profile

The **NFW profile** (Navarro-Frenk-White) from N-body simulations:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \quad (39)$$

where r_s is the scale radius and ρ_s is a characteristic density.

8.3 DM Candidates

Candidate	Mass range	Production	Status
WIMPs	$\sim 10 \text{ GeV} - 10 \text{ TeV}$	Freeze-out	Constrained
Axions	$\sim \mu\text{eV} - \text{meV}$	Misalignment	Active search
Sterile neutrinos	keV	Mixing	Constrained
Primordial BHs	$10^{-17} - 10^2 M_\odot$	Gravitational	Open window

8.4 WIMP Miracle and Freeze-Out

A WIMP with mass m_χ and annihilation cross section $\langle\sigma v\rangle$ achieves the correct relic density through **thermal freeze-out**. The Boltzmann equation:

$$\dot{n}_\chi + 3Hn_\chi = -\langle\sigma v\rangle (n_\chi^2 - n_{\text{eq}}^2) \quad (40)$$

At freeze-out temperature $T_f \approx m_\chi/20$, the relic abundance:

$$\Omega_\chi h^2 \approx \frac{0.1 \text{ pb}}{\langle\sigma v\rangle} \quad (41)$$

Remarkably, electroweak-scale cross sections naturally give $\Omega_\chi \sim 0.3$ — the **WIMP miracle**.

9 Dark Energy

9.1 Observational Discovery

In 1998, Type Ia supernovae observations (Perlmutter et al.; Riess et al.) revealed the universe's expansion is **accelerating**: $\ddot{a} > 0$, requiring $w < -1/3$.

9.2 The Cosmological Constant

The simplest model: $\Lambda = \text{const}$, with $w_\Lambda = -1$. The **cosmological constant problem**: quantum field theory predicts:

$$\rho_{\text{vac}}^{\text{QFT}} \sim M_{\text{Pl}}^4 \approx 10^{74} \text{ GeV}^4 \quad (42)$$

while observation gives $\rho_\Lambda \approx 10^{-47} \text{ GeV}^4$ — a discrepancy of **120 orders of magnitude**.

9.3 Dynamic Dark Energy

Alternatives to Λ include:

Quintessence: A rolling scalar field ϕ with:

$$w_\phi = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} \quad (43)$$

For $\dot{\phi}^2 \ll V$: $w_\phi \rightarrow -1$. Equations of motion same as inflaton.

Phantom energy: $w < -1$, leads to eventual *Big Rip*.

$w_0 w_a$ **parameterisation**:

$$w(a) = w_0 + w_a(1 - a) \quad (44)$$

Current constraints: $w_0 \approx -0.95 \pm 0.08$, $w_a \approx -0.5 \pm 0.5$ (DESI 2024).

10 Structure Formation

10.1 Linear Perturbation Theory

Small inhomogeneities are described by density contrast $\delta \equiv (\rho - \bar{\rho})/\bar{\rho}$. In Fourier space, the linearised equations give:

Growth Equation for Matter Perturbations

$$\ddot{\delta}_k + 2H\dot{\delta}_k - \frac{3}{2}H_0^2\Omega_m a^{-3}\delta_k = 0 \quad (45)$$

The growing mode solution during matter domination: $\delta \propto a$.

10.2 Transfer Function and Matter Power Spectrum

Modes entering the horizon during radiation domination are suppressed. The **transfer function** $T(k)$ encodes this:

$$P(k, z) = A_s k^{n_s} T^2(k) D^2(z) \quad (46)$$

where $D(z)$ is the **linear growth factor**:

$$D(z) \propto H(z) \int_z^\infty \frac{(1+z')dz'}{H^3(z')} \quad (47)$$

10.3 Jeans Analysis

Perturbations grow only above the **Jeans length**:

$$\lambda_J = c_s \sqrt{\frac{\pi}{G\bar{\rho}}} \quad (48)$$

Modes with $\lambda < \lambda_J$ oscillate (pressure-supported); modes $\lambda > \lambda_J$ grow.

10.4 N-Body Simulations

Non-linear structure ($\delta \gtrsim 1$) is modelled numerically. The **halo mass function** dn/dM (Press-Schechter theory):

$$\frac{dn}{dM} = \sqrt{\frac{2}{\pi}} \frac{\bar{\rho}}{M} \frac{\delta_c}{\sigma^2} \left| \frac{d\sigma}{dM} \right| \exp\left(-\frac{\delta_c^2}{2\sigma^2}\right) \quad (49)$$

where $\delta_c \approx 1.686$ is the linear collapse threshold and $\sigma^2(M)$ is the variance in the linear density field.

10.5 Baryon Acoustic Oscillations (BAO)

The acoustic sound horizon $r_s \approx 147$ Mpc imprints a **standard ruler** in the matter power spectrum, visible as a peak at $k \approx 0.06$ h/Mpc and in galaxy correlation functions at separation $r \approx 150$ Mpc.

11 Gravitational Waves in Cosmology

11.1 Tensor Perturbations

Gravitational waves in GR satisfy:

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2 h_{ij}}{a^2} = 0 \quad (50)$$

In the transverse-traceless gauge, two polarisation states h_+ , $h_\times$.

11.2 Stochastic Gravitational Wave Background (SGWB)

A **stochastic gravitational wave background** is predicted from:

- Inflationary quantum fluctuations (scale-invariant spectrum)
- Phase transitions in the early universe
- Cosmic string networks
- Binary compact object mergers

Characterised by the energy density spectrum:

$$\Omega_{\text{GW}}(f) = \frac{f}{\rho_c} \frac{d\rho_{\text{GW}}}{df} \quad (51)$$

PTA (Pulsar Timing Array) experiments including NANOGrav, PPTA, and EPTA reported evidence for a SGWB signal in 2023, potentially of astrophysical or cosmological origin.

12 Quantum Cosmology and the Planck Era

12.1 Planck Scale

At energy $E_{\text{Pl}} = \sqrt{\hbar c^5/G} \approx 1.22 \times 10^{19}$ GeV, quantum gravitational effects dominate. Natural units:

$$\ell_{\text{Pl}} = \sqrt{\frac{\hbar G}{c^3}} \approx 1.6 \times 10^{-35} \text{ m}, \quad t_{\text{Pl}} \approx 5.4 \times 10^{-44} \text{ s} \quad (52)$$

12.2 Wheeler-DeWitt Equation

Canonical quantisation of GR yields the **Wheeler-DeWitt equation**:

$$\hat{\mathcal{H}} \Psi[g_{ij}] = 0 \quad (53)$$

where $\Psi[g_{ij}]$ is the **wave function of the universe** (a functional of 3-geometries). In minisuperspace (FLRW):

$$\left[-\frac{\partial^2}{\partial a^2} + U(a) \right] \Psi(a) = 0 \quad (54)$$

12.3 Hartle-Hawking No-Boundary Proposal

The **no-boundary wave function** (Hartle-Hawking 1983):

$$\Psi_{\text{HH}} = \int_{\text{compact}} \mathcal{D}g \mathcal{D}\phi e^{-S_E[g,\phi]} \quad (55)$$

integrates over compact Euclidean 4-geometries with no boundary — a “universe created from nothing.”

12.4 Loop Quantum Cosmology

Loop Quantum Gravity (LQG) quantises area in units of ℓ_{Pl}^2 . Applied to cosmology, it replaces the Big Bang singularity with a **Big Bounce**: the universe contracts, reaches minimum volume $V_{\text{min}} \sim \ell_{\text{Pl}}^3$, then re-expands. The effective Friedmann equation becomes:

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_c^{\text{LQC}}} \right) \quad (56)$$

where $\rho_c^{\text{LQC}} \approx 0.41 \rho_{\text{Pl}}$.

13 Current Research Frontiers

13.1 The Hubble Tension

The Hubble Tension — $\sim 5\sigma$ Discrepancy

- **Early universe** (Planck CMB): $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$
- **Late universe** (SH0ES Cepheids+SNe Ia): $H_0 = 73.0 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$

Proposed solutions include: early dark energy, interacting dark sectors, modified recombination, new relativistic species (ΔN_{eff}), or systematic errors.

13.2 The S_8 Tension

Weak lensing surveys (KiDS, DES, HSC) measure the matter clustering parameter:

$$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3} \quad (57)$$

at $\sim 2\text{--}3\sigma$ lower than Planck predictions. May indicate new dark matter microphysics or baryonic feedback uncertainties.

13.3 DESI and Dark Energy

The Dark Energy Spectroscopic Instrument (DESI, 2024–2025) mapped ~ 40 million galaxies. BAO measurements suggest $w_0 > -1$ and $w_a < 0$, hinting that dark energy may be **dynamical** at $\sim 3\sigma$ significance — a potential revolution if confirmed.

13.4 21-cm Cosmology

Neutral hydrogen emits/absorbs at $\lambda = 21$ cm. The **21-cm signal** probes the **Epoch of Reionization** ($6 \lesssim z \lesssim 20$) and the Dark Ages ($z \sim 20$ –200). HERA, SKA, and EDGES are key experiments. EDGES (2018) detected an anomalously deep absorption trough at $z \approx 17$ possibly indicating dark matter-baryon interactions.

13.5 Gravitational Wave Cosmology

Standard sirens: Binary mergers from LIGO/Virgo provide independent H_0 measurements via:

$$d_L \propto \frac{\mathcal{M}^{5/6}}{h}, \quad \mathcal{M} = (m_1 m_2)^{3/5} / (m_1 + m_2)^{1/5} \quad (58)$$

LISA (launch ~ 2035) will probe massive black hole mergers at $z \sim 20$ and the SGWB.

13.6 Primordial Black Holes

Primordial Black Holes (PBHs) formed from density fluctuations in the early universe could comprise part of dark matter. Constraints from:

- Hawking radiation (evaporation): $M_{\text{PBH}} \gtrsim 10^{15} \text{ g}$
- Microlensing (MACHO, EROS, Subaru HSC)
- GW mergers (LIGO)
- CMB spectral distortions

A mass window 10^{-16} – $10^{-11} M_\odot$ remains open.

13.7 Inflation Model Selection

Next-generation CMB experiments (CMB-S4, LiteBIRD, Simons Observatory) aim to:

- Detect or rule out $r \gtrsim 0.003$ (B-mode polarisation)
- Measure n_s to 0.002 precision
- Constrain primordial non-Gaussianity f_{NL} to $\sigma(f_{\text{NL}}) \sim 1$

13.8 Neutrino Cosmology

The cosmic neutrino background ($C\nu B$) has $T_\nu = (4/11)^{1/3}T_\gamma \approx 1.95$ K.

The **effective number of relativistic species**:

$$N_{\text{eff}} = 3.044 \pm 0.002 \quad (\text{SM prediction}) \quad (59)$$

Neutrino mass sum $\sum m_\nu < 0.12$ eV (Planck 2018). KATRIN and PTOLEMY aim to probe absolute neutrino mass. Cosmological observations are competitive with $\sigma(\sum m_\nu) \sim 0.02$ eV forecast for future surveys.

13.9 Multiverse and Eternal Inflation

Many inflationary potentials lead to **eternal inflation**: inflation never ends globally — bubble nucleation creates causally disconnected “pocket universes.” The **string landscape** ($\sim 10^{500}$ vacua) provides a multiverse of different physical constants. The **measure problem** (how to assign probabilities over the multiverse) remains unsolved.

14 The Standard Model of Cosmology: Λ CDM

14.1 Parameters and Predictions

Λ CDM is described by 6 independent parameters (Planck 2018):

Parameter	Description	Value
$\Omega_b h^2$	Baryon density	0.02237 ± 0.00015
$\Omega_c h^2$	Cold DM density	0.1200 ± 0.0012
θ_s	Angular sound horizon	1.04092 ± 0.00031
τ	Optical depth (reionisation)	0.054 ± 0.007
$\ln(10^{10} A_s)$	Scalar amplitude	3.044 ± 0.014
n_s	Spectral index	0.9649 ± 0.0042

14.2 Timeline of the Universe

Epoch	Time	Temperature
Planck epoch	$< 10^{-43}$ s	$> 10^{32}$ K
Inflation	10^{-36} – 10^{-32} s	$\sim 10^{28}$ K
Baryogenesis	$\sim 10^{-10}$ s	$\sim 10^{15}$ K
QCD transition	$\sim 10^{-5}$ s	$\sim 10^{12}$ K
BBN	3 min – 20 min	10^{10} – 10^9 K
Matter-radiation equality	50,000 yr	9500 K
Recombination	380,000 yr	3000 K
Reionisation	~ 400 Myr	~ 40 K
Today	13.8 Gyr	2.725 K

15 Conclusion

Modern cosmology has achieved a precise, predictive standard model (Λ CDM) grounded in general relativity, quantum field theory, and thermodynamics. Yet profound mysteries remain:

Open Questions in Cosmology

1. **What is dark matter?** No particle has been detected despite decades of searches.
2. **What is dark energy?** The cosmological constant problem is the most severe fine-tuning problem in physics.
3. **What drove inflation?** The inflaton is unidentified; r is undetected.
4. **Why is there matter (not antimatter)?** Baryogenesis requires C , CP violation and departure from thermal equilibrium.
5. **What happened before/at the Big Bang?** A quantum theory of gravity is needed.
6. **What is the fate of the universe?** Heat death, Big Rip, or Big Crunch?
7. **Is the Hubble tension pointing to new physics?** If systematic errors are excluded, new cosmological models are needed.

The next decade — with experiments including DESI, CMB-S4, Euclid, LISA, SKA, and the Vera Rubin Observatory — promises to sharpen or shatter our current paradigm, potentially revealing entirely new physics at the intersection of the very large and the very

small.

*“To make an apple pie from scratch,
you must first invent the universe.”*

— Carl Sagan